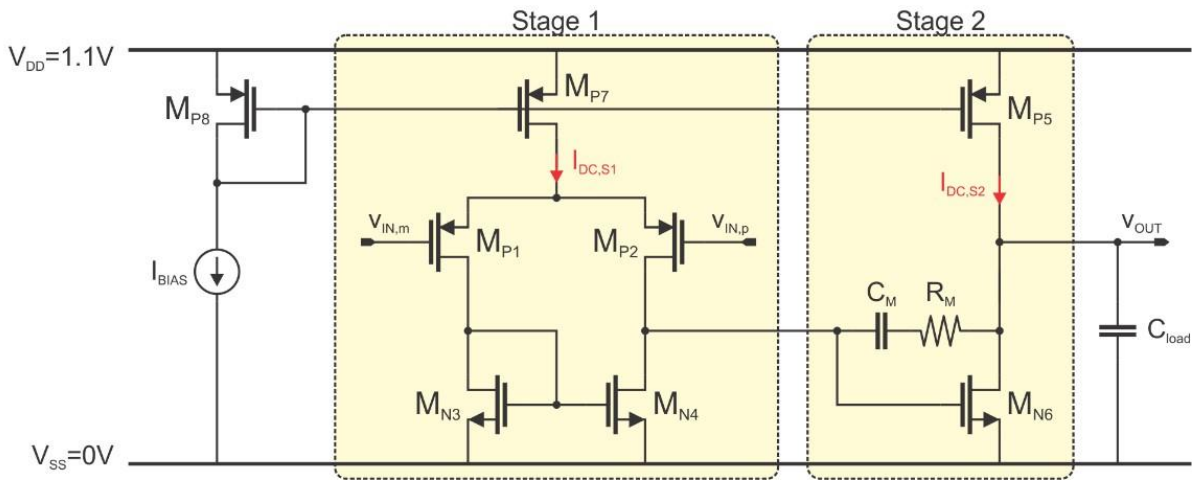
Group data

Group number	7
Name – Student 1	[redacted]
Name – Student 2	[redacted]

Goal:



Design of the 2stage OpAmp such that it passes the given specifications (insert your specs below):

$C_{load}[pF]$	20
DC gain [dB]	50
f_{GBW} [MHz]	120
Phase margin (PM) [deg]	> 70
Output swing [V]	> 0.7

Plan: Design of the 2-stage OpAmp on paper (10 points)

1) Calculate the V_{OV},g_m, I_{DS} and L of each transistor (see exceptions under "*Remark*") and the required C_M and R_M in the OpAmp circuit. For that, insert all your calculations as well as your $\frac{g_m}{g_{ds}}$, $\frac{g_m}{I_D}$ plots you used for your handcalculations below:

Hints:

- *First, plot $g^g ds^m$ and $^g ID^m$ across V_{OV} and gatelength L and do it again across V_{GS}, as presented in the 1st session. Think about whether to create the plots for a PMOS or for a NMOS device.*
- *Then, based on those plots, start your calculations.*
- *Furthermore, you can assume the following:*
  (1) the OpAmp is designed in triple-well-technology (i.e. $V_{SB} = 0$); (2) $V_{dsat} \approx V_{OV}$ and $V_{dsat} \geq 50mV$ across V_{OV}; (3) $C_M = \frac{C_L}{4}$

Remarks:

- *For Mp5, Mp7 and Mp8 you are not required to calculate the g_m*

Plots used for the handcalculations:

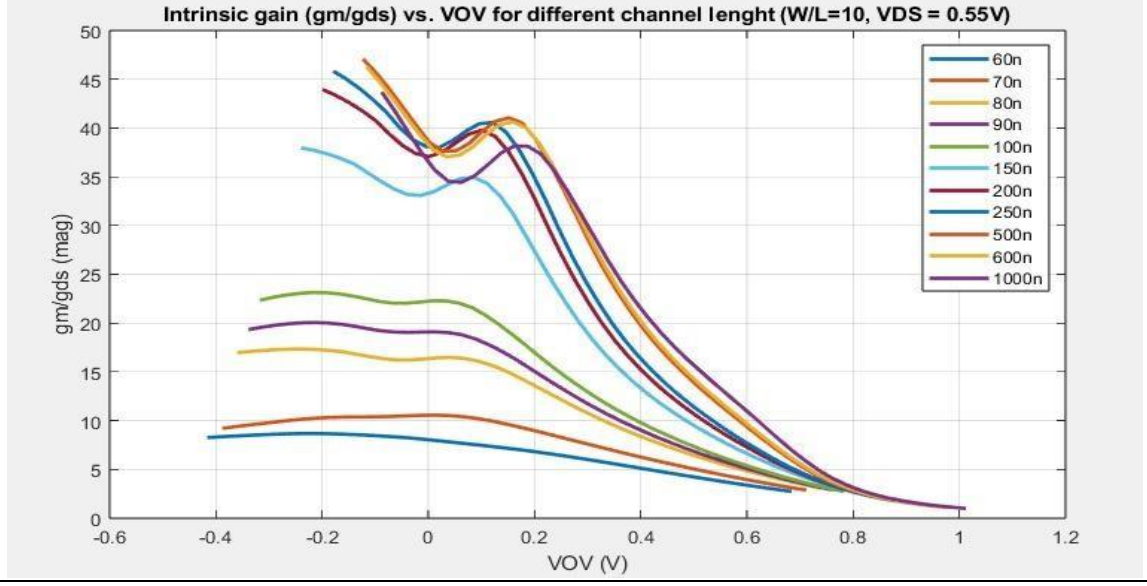


Figure 1: $\frac{g_m}{g_{ds}}$ vs V_{OV} across gatelength L (NMOS)

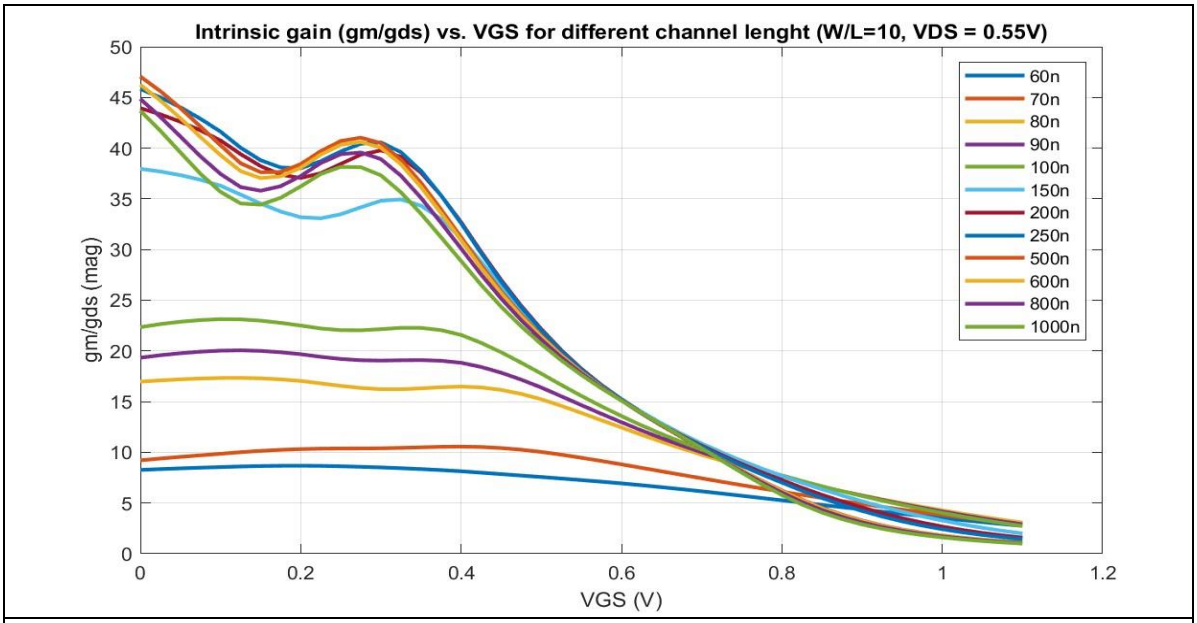


Figure 2: $\frac{g_m}{g_{ds}}$ vs V_{GS} across gatelength L (NMOS)

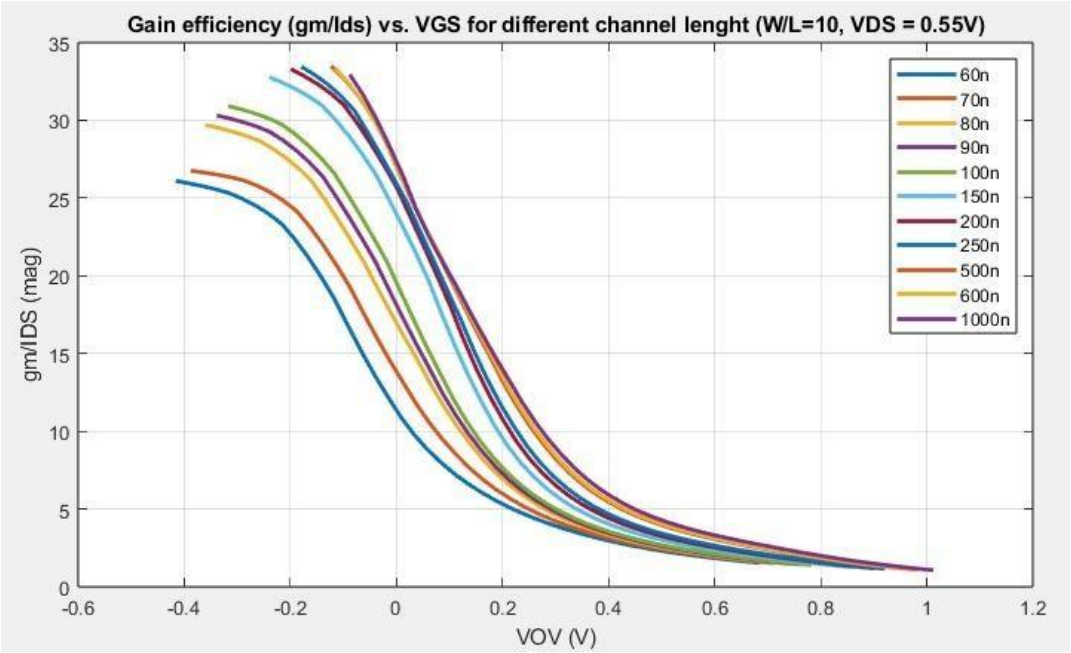


Figure 3: $\frac{g_m}{I_D}$ vs V_{OV} across gatelength L (NMOS)

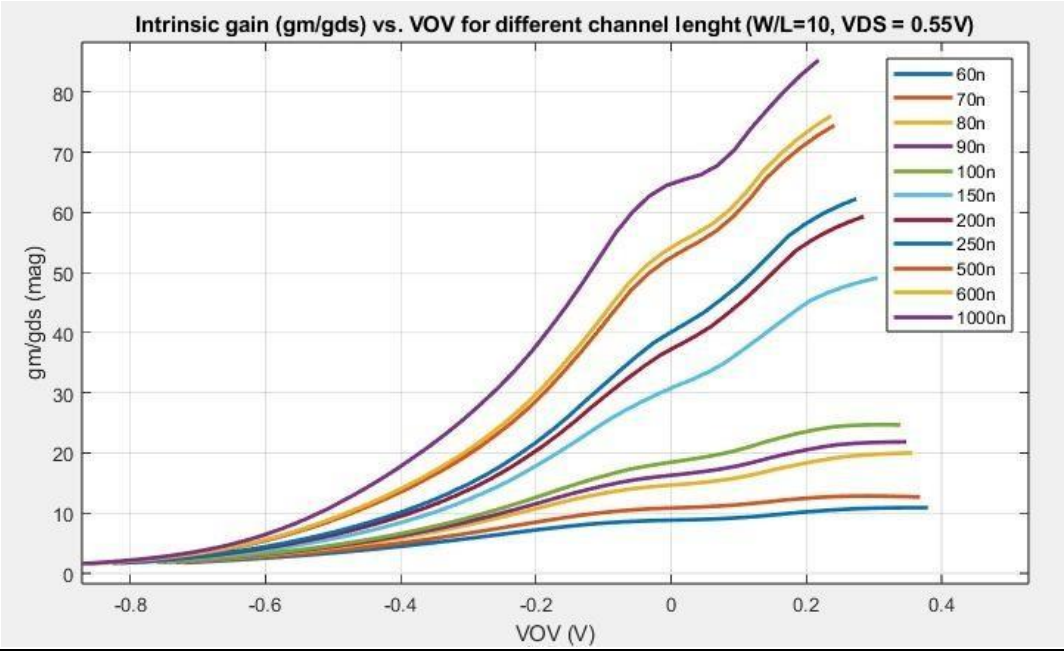


Figure 4 : $\frac{g_m}{g_{ds}}$ vs V_{OV} across gatelength L (PMOS)

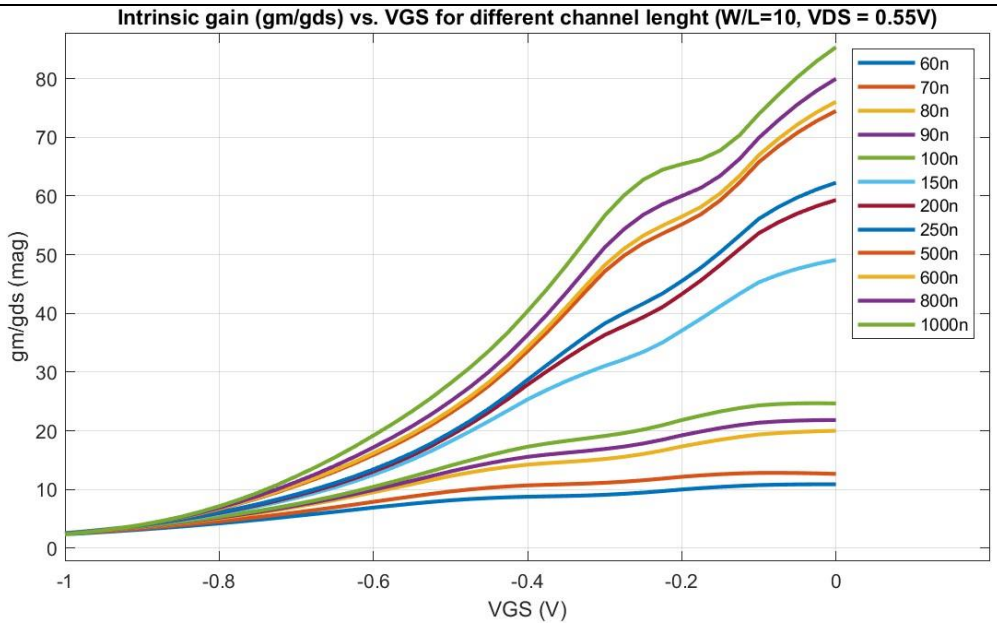


Figure 5: $\frac{g_m}{g_{ds}}$ vs V_{GS} across gatelength L (PMOS)

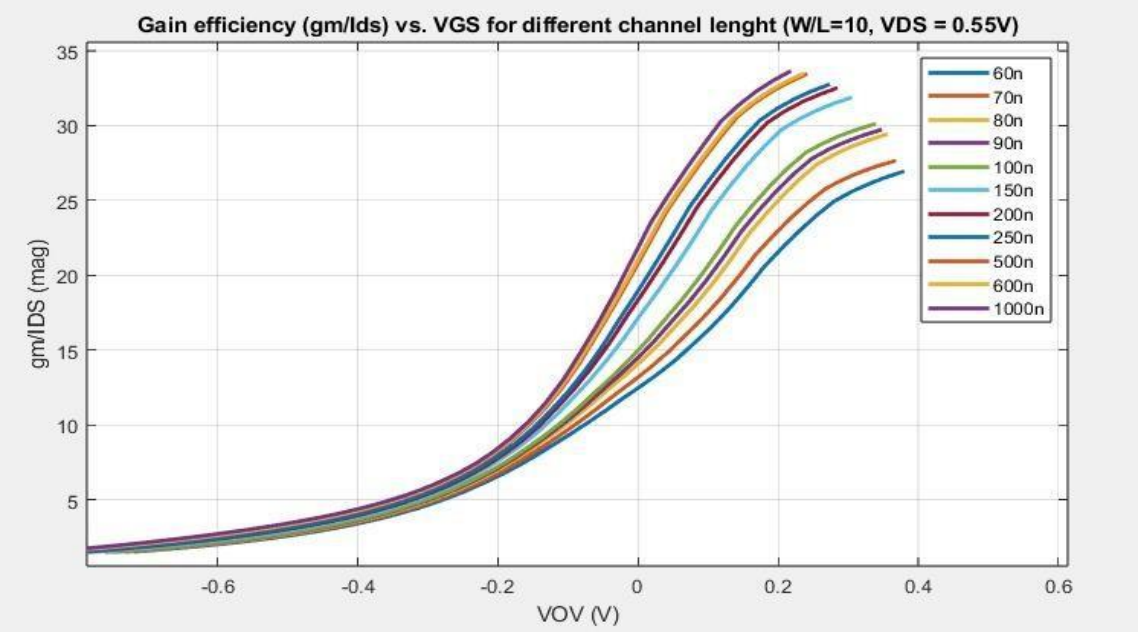


Figure 6: $\frac{g_m}{I_D}$ vs V_{OV} across gatelength L (PMOS)

1 Key Devices

Stage1: A differential pair (transistors M_{P1},M_{P2},M_{N3},M_{N4},M_{P7}) providing the first voltage gain.
Stage2: A common-source gain stage (transistors M_{P5}, M_{N6}) with Miller compensation network C_M and resistor R_M. The load capacitor is denoted by C_{load}.

The bias current sources are provided by transistors M_{P8}, and additional current mirror devices ensuring the required tail current and proper operation of each stage.

2 Small-Signal Analysis and Gain Equations

2.1 DC Gain of Each Stage

The OTA features two gain stages. The first stage is based on an NMOS differential pair with folded cascode loads, and the second stage is an NMOS common-source amplifier.

2.2 First-Stage Gain

Let g_{m1} be the transconductance of the input differential pair transistor (M_{P1}), and let g_{o1} and g_{o3} be the small-signal output conductances of M_{P1} and M_{N3}. The DC gain of the first stage is

$$A_{v1} = -\frac{g_{m1}}{g_{o1} + g_{o3}}. \tag{1}$$

2.3 Second-Stage Gain

In the second stage, transistor M_{N6} provides the transconductance, while transistors M_{P5} and M_{N6} contribute to the output conductance. Denote g_{m6} as the transconductance of M_{N6}, and g_{o5} and g_{o6} the output conductances of M_{P5} and M_{N6}, respectively. The DC gain of the second stage is

$$A_{v2} = -\frac{g_{m6}}{g_{o5} + g_{o6}}. \tag{2}$$

2.4 Overall DC Gain

The overall DC gain of the OTA is

$$A_{v,DC} = A_{v1}\,A_{v2} = \left(\frac{-g_{m1}}{g_{o1} + g_{o3}}\right) \times \left(\frac{-g_{m6}}{g_{o5} + g_{o6}}\right). \tag{3}$$

The design target of ≈ 316.228 in magnitude is satisfied by sizing the transistors such that the product of the two gains meets the requirement.

2.5 Noise Analysis

The gain of the second stage was chosen to be significantly lower than the first stage to reduce the impact of its noise on the overall performance. This configuration ensures that the noise contribution of the second stage is minimized relative to the amplified signal from the first stage.

3 Pole and Zero Analysis

3.1 Poles

The small-signal transfer function exhibits up to three relevant poles in the design shown, labeled p , p_2, p_3.

In particular:

1. p_1 arises from the diode-connected transistor.

$$P_1 = -\frac{g_{m3}+g_{ds1}+g_{ds3}}{C_{db1}+C_{gd1}+2C_{gs3}+2C_{gb3}+C_{gd3}} \approx -\frac{g_{m3}}{C_{n1}} \quad (4)$$

2. p_2 is the dominant pole once Miller compensation is applied. The expressions show the splitting effect:

$$P_2 = P_d = -\frac{(g_{ds1}+g_{ds3})}{C_M(1-A_{v2})} = -\frac{(g_{ds1}+g_{ds3})}{C_M\left(1+\frac{g_{m6}}{g_{ds5}+g_{ds6}}\right)} \quad (5)$$

3. p_3 is the output node pole:

$$p_3 \approx -\frac{g_{m6}+g_{o5}+g_{o6}}{C_L} \approx -\frac{g_{m6}}{C_L}. \quad (6)$$

3.2 Zeros

Giving the fact that one signal path from the output goes through a pole, but the other does not go through a pole, there is a pole-zero doublet at node 1 such that:

$$z_1 = 2p_1 \quad (7)$$

Introducing the Miller capacitance creates a negative zero that can further decrease the phase margin. We make the output node a virtual ground. The drain current is equal to $g_{m6}V_{Gs6}$and the capacitive branch current is $\frac{V_{Gs6}}{R_m+\frac{1}{C_M s}}$, and these two currents are equal to each other.

$$g_{m6} = \frac{1}{R_m+\frac{1}{C_M s}} \quad => \quad z_2 = \frac{1}{C_M\left(\frac{1}{g_{m6}}-R_M\right)} \quad (8)$$

4 Calculations based on design parameters

4.1 p_3 based on PM

To achieve a phase margin greater than 70°, we utilize the graph between $\frac{|secend\ dominant\ pole|}{\omega_{GBW}}$ vs PM. This relation implies that the farther the dominant pole is from the origin, the higher the achievable phase margin. However, due to pole splitting introduced by the Miller capacitance, placing the second dominant pole (p_{2nd}) too high would result in an undesirably too low dominant

pole. This would lead to a lower gain-bandwidth product (f_{GBW}). Therefore, it is necessary to strike a balance.

Let us consider a phase margin of PM = 76°.

$$PM = 76^{\circ}.\ PM = 90 - tg^{-1}\left(\frac{GBW}{p_3}\right) ==> \quad \frac{|p_3|}{\omega_{GBW}} = 3.8916 \tag{9}$$

We can compute $|p_3|$ as follows:

$$|p_3| = 3.8916 \cdot \omega_{GBW} = 3.8916 \cdot 2\pi \cdot f_{GBW} \tag{10}$$

To achieve a the required f_{GBW}, let us consider a higher f_{GBW} = 120MHz

Substituting the values, we find:

$$|p_3| = 3.8916 \times 2\pi \times 120 \times 10^6 = 2934.2 \times 10^6\ rad/s \tag{11}$$

4.2 gm1 Based on fGBW

The relationship between the gain-bandwidth product and the first stage transconductance g_{m1} with the Miller capacitor C_M is:

$$f_{GBW} = A_v\, f_{p_{dominant}} = |A_{v1}\, A_{v2}|\, f_{p2} \tag{13}$$

$$\approx |A_{v1}\, A_{v2}| \cdot \frac{g_{o1} + g_{o3}}{2\,\pi\, C_M\, |A_{v2}|} \tag{14}$$

$$= |A_{v1}|\, \frac{g_{o1} + g_{o3}}{2\,\pi\, C_M} \tag{15}$$

$$= \frac{g_{m1}}{g_{o1} + g_{o3}} \cdot \frac{g_{o1} + g_{o3}}{2\,\pi\, C_M} \tag{16}$$

$$= \frac{g_{m1}}{2\,\pi\, C_M}. \tag{17}$$

$$g_{m1} = 2\pi\, C_M f_{GBW} = 2\pi \times 5 \times 10^{-12} \times 120 \times 10^6 \quad \Rightarrow \quad g_{m1} \approx 3768\ \mu s \tag{18}$$

4.3 C_M

The choice of $C_M \approx 5pF$ originates from the requirement that

$$C_M = \frac{1}{4} C_L = \frac{20}{4} = 5\ PF \tag{19}$$

4.4 gm6

$$p_3 \approx -\frac{g_{m6}}{C_L} \tag{20}$$

Since we require

$$p_3 = -2934.2M\ rad/s \quad \Rightarrow \quad g_{m6} = -C_L \times p_3 = 20pF \times -2934.2M\ rad/s \tag{21}$$

$$\Rightarrow \qquad g_{m6} = 58644\mu s \qquad (22)$$

Of course, there is a rule of thumb in this circuit as follows:

$$\begin{cases} PM > 60^{\circ} & => \quad g_{m6} \geq 10 g_{m1} \\ PM > 70^{\circ} & => \quad g_{m6} \geq 15 g_{m1} \end{cases} \qquad (23)$$

4.5 R_M

Choose R_M so that the unwanted LHP zero z_2 directly cancels an existing pole(p_3) in the system, which can lead to a phase margin of approximately 90°:

$$-\frac{g_{m6}}{C_L} = \frac{1}{C_M\left(\frac{1}{g_{m6}} - R_M\right)} \qquad => \qquad R_M = \frac{1}{g_{m6}}\left(\frac{C_L + C_M}{C_M}\right) \qquad (24)$$

$$R_M = \frac{1}{58644 \times 10^{-6}}\left(\frac{20+5}{5}\right) = 85.26\Omega$$

5 Transistor Bias and Operating-Region Selection

5.1 Saturation Requirements

All transistors must operate in saturation for the small-signal analysis and gain derivations to be valid.

For an NMOS transistor in saturation, the following condition applies:

$$V_{DS} > V_{GS} - V_{Th} > 0 \qquad (25)$$

For a PMOS device, the inequality has the appropriate sign inversion:

$$V_{DS} < V_{GS} - V_{Th} < 0 \qquad (26)$$

5.2 Voltage Swing Considerations

Output Voltage Swing:

$$V_{out,max} = V_{DD} - |V_{OV,P5}| \quad , \quad V_{out,min} = |V_{OV,N6}|.$$

Hence, the total swing is

$$V_{swing} = V_{out,max} - V_{out,min} = V_{DD} - \left(|V_{OV,P5}| + |V_{OV,N6}|\right).$$

Headroom Constraints: We choose V_{OV} values such that

$$V_{swing} > 0.7V \Rightarrow \quad |V_{OV.P5}| + |V_{OV.N6}| < 0.4 \ \ V$$

The output bias point is set so that V_{out} rests near $\frac{V_{DD}}{2}$ when there is no signal, ensuring symmetric

$$V_{out} = V_{DS.N6} = \frac{V_{DD}}{2} = \frac{1.1}{2} \quad => \quad V_{DS.N6} = 0.55\ V \qquad (27)$$

$$V_{DS.P5} = V_{DS.N6} - V_{DD} = 0.55 - 1.1 \quad => \quad V_{DS.P5} = -0.55\ V \qquad (28)$$

This allows us to make the required plots necessary to determined the other parameters.

5.3 Overall Design Strategy

The design goal is to balance power consumption, gain, and output swing. Throughout the amplifier stages, we have chosen to bias most transistors in moderate inversion. This region provides a favorable trade-off between:

Transconductance efficiency (g_m/I_D) characteristic of weak inversion,

Reasonable drive capability and speed characteristic of strong inversion.

Additionally, we keep certain devices at relatively large channel length (L) to reduce channel-length modulation and improve their performance as current sources.

5.4 Second-Stage Devices

In the second gain stage:

- **Transistor M_{N6} (Common-Source Amplifier):**
  - Should have a low intrinsic gain so that the gain of the second stage is minimized, therefore the channel length should be small
  - Operates in moderate inversion to have stable current

Therefore, the choices made were:

$$V_{OV.N6} = 0 \quad . \quad L_6 = 100nm \tag{29}$$

Based on the plots made:

$$(V_{OV,N6} = 0,\ L_6 = 100\,nm) \implies \frac{g_{m6}}{g_{ds6}} = 22.238 \implies g_{ds6} = 2637\ \mu s \tag{30}$$

$$(V_{OV,N6} = 0,\ L_6 = 100\,nm) \implies \frac{g_{m6}}{g_{ds6}} = 22.238 \implies V_{GS.N6} = 0.31\ V \tag{31}$$

$$(V_{OV,N6} = 0,\ L_6 = 100\,nm) \implies \frac{g_{m6}}{I_{ds6}} = 19.555 \implies I_{ds6} = 2999\ \mu A \tag{32}$$

Transistor M_{P5} (Current Source):

-Implements the active load for the second stage.

-A longer L is chosen to reduce channel-length modulation and maintain a stable, well-defined current.

-Biased in moderate inversion, giving a decent trade-obetween the current and power.

Therefore, the choices made were:

(33)

$$V_{OV.P5} = -0.2 \quad . \quad L_5 = 1000nm$$

Because M_{P5} and M_{N6} are in series, they share the same current. Therefore,

$$I_{ds5} = I_{ds6} = I_{DC.S2} \quad => \quad I_{DC.S2} = 2999\ \mu A \tag{34}$$

Based on the plots made:

$$(V_{OV,P5} = -0.2\,V,\ L_5 = 1000\,nm) \implies \frac{g_{m5}}{I_{ds5}} = 8.4442 \implies g_{m5} = 25324\ \mu s \tag{35}$$

$$(V_{OV,P5} = -0.2\,V,\ L_5 = 1000\,nm) \implies \frac{g_{m5}}{g_{ds5}} = 37.79 \implies g_{ds5} = 670\ \mu s \tag{36}$$

$$(V_{OV,P5} = -0.2\,V,\ L_5 = 1000\,nm) \implies \frac{g_{m5}}{g_{ds5}} = 37.79 \implies V_{GS.P5} = -0.4\,V \tag{37}$$

Computing the gain of the second stage A_{v2}

$$A_{v2} = -\frac{g_{m6}}{g_{o5} + g_{o6}} = -\frac{58644}{670 + 2637} = -17.733$$

5.5 Stage 1 Devices

First Analysis

- Since M_{N3} is diode connected, and because of the current mirrors:

$$V_{GS.N6} = V_{GS.N4} = V_{GS.N3} = V_{DS.N3} = V_{DS.N4} = 0.31 \tag{39}$$

- For M_{P8} and the current mirrors :

$$V_{GS.P5} = V_{GS.P7} = V_{GS.P8} = -0.4 \tag{40}$$

- For simplicity, let us consider that M_{P8} (diode-connected) and M_{P7} are identical such that we can consider the following approximation

$$V_{GS.P8} = V_{DS.P8} = V_{GS.P7} = V_{DS.P7} = -0.4 \tag{41}$$

- To ensure symmetry in therst-stage differential ampli er, M_{P1} and M_{P2} must be identical, as must M_{N3} and M_{N4}. Therefore:

$$V_{OV.P1} = V_{OV.P2} \quad . \quad g_{m1} = g_{m2} \quad . \quad g_{01} = g_{o2} \tag{42}$$

$$V_{OV.N3} = V_{OV.N4} \quad . \quad g_{m3} = g_{m4} \quad . \quad g_{03} = g_{o4} \tag{43}$$

This ensures balanced operation, improves common-mode rejection, and make the calculations easier.

Differential-amplifier Transistors M_{P1},M_{P2}

- The inversion level chosen was moderate inversion to ensure a good trade off between the intrinsic gain and good current
- Knowing that, $A_{v2} \approx -18$. By trial and error, we found a length of 500*nm* is a good compromise for the limiting the channel length modulation. Therefore:

$$V_{OV.P1} = V_{OV.P2} = 0 \quad . \quad L_1 = L_2 = 500\ nm \tag{44}$$

Based on the plots:

$$(V_{OV,P1} = 0\,V,\ L_1 = 500\,nm) \implies \frac{g_{m1}}{g_{ds1}} = 52.51 \implies V_{GS.P1} = -0.2415\ V \tag{45}$$

$$(V_{OV,P1} = 0\,V,\ L_1 = 500\,nm) \implies \frac{g_{m1}}{g_{ds1}} = 52.51 \implies g_{ds1} = 71.76\ \mu s \tag{46}$$

$$(V_{OV,P1} = 0\,V,\ L_1 = 500\,nm) \implies \frac{g_{m1}}{I_{ds1}} = 20.92 \implies I_{ds1} = 180\ \mu A \tag{47}$$

- **Current-Source and Current-mirror Transistors M_{P8},M_{P7},M_{N3},M_{N4}:**

These transistors form the various current mirrors and tail-current sources throughout the amplier:

- Longer Channel Length: By choosing L larger than the minimum, we reduce the output conductance g_o and thus improve current-source quality.
- Moderate Inversion Bias: The overdrive voltage is selected to be modest to balance current efficiency with acceptable device sizes.
- Power Budget: Because these transistors set or mirror bias currents, we do not want excessively large drain current, and sized just large enough for matching while meeting headroom (VDS≥VDsat) so the power budget is respected.

To have good current sources, we take

$$L_7 = L_8 = 1000\ nm$$

Based on the plots:

$$(V_{GS,P8} = -0.4\,V,\ L_8 = 1000\,nm) \implies \frac{g_{m8}}{g_{ds8}} = 37.79 \tag{49}$$

$$=> \quad V_{OV.P7} = V_{OV.P8} = -0.2 \tag{50}$$

$$(V_{OV,P8} = -0.2\,V,\ L_8 = 1000\,nm) \implies \frac{g_{m8}}{I_{ds8}} = 8.4442 \tag{51}$$

We choose a current mirror with a ratio of 1:1 for M_{P8} and M_{P7}. Therefore:

$$I_{ds8} = I_{ds7} = I_{DC.S1} = I_{bias} = 2I_{ds1} \quad => \quad I_{bias} = I_{DC.S1} = 360\ \mu A \tag{52}$$

By trial and error:

$$L_3 = L_5 = 500\ nm \tag{53}$$

We also know that:

$$Ids3 = Ids1 = Ids2 = Ids4 \tag{54}$$

Based on the plots:

$$(V_{GS,N3} = 0.31\,V,\ L_3 = 500\,nm) \implies \frac{g_{m3}}{g_{ds3}} = 39.84 \implies \quad V_{OV.N3} = 0.195\ V \tag{55}$$

$$(V_{OV,N3} = 0.195\,V,\ L_3 = 500\,nm) \implies \frac{g_{m3}}{I_{ds3}} = 13.57 \implies \quad g_{m3} = 2442.6\ \mu s \tag{56}$$

$$(V_{OV,N3} = 0.195\,V,\ L_3 = 500\,nm) \implies \frac{g_{m3}}{g_{ds3}} = 39.84 \implies \quad g_{ds3} = 61.31\ \mu s \tag{57}$$

- Computing the gain of the first stage A_{v1}

$$A_{v1} = -\frac{g_{m1}}{g_{o1}+g_{03}} = -\frac{3768}{61.31+71.76} = -28.316 \tag{58}$$

5.6 Total DC Gain

$$A_v = A_{v1}A_{v2} = -28.316 \times -17.733 \Rightarrow \quad A_v = 502.128 = 54\ dB \tag{59}$$

Place the obtained values in the table below. Only the values in here need to be calculated / chosen in the handcalculations. Each value should be chosen with a reason which you have to provide above.

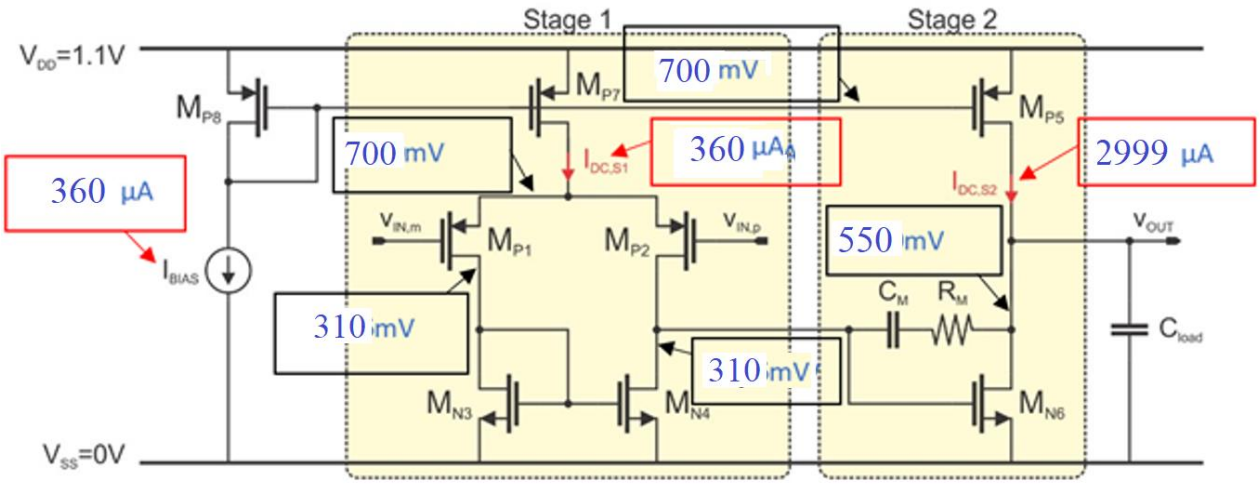
Device	L [nm]	I_{ds} [µA]	Vov [V]	g_m [µS]
M_{p1}	500	180	0	3768
M_{p2}	500	180	0	3768
M_{n3}	500	180	0.195	2442.6
M_{n4}	500	180	0.195	2442.6
M_{p5}	1000	2999	-0.2	
M_{n6}	100	2999	0	58644
M_{p7}	1000	360	-0.2	
M_{p8}	1000	360	-0.2	

Device	Units	Value
R_M	Ω	85.26
I_{BIAS}	µA	360

Place all the calculated voltages (in the black box) and all currents (in the red box) on the circuit depicted below and calculate also (include formula):

2)

$V_{cm,in,min}$	$V_{GS,3} - V_{Dsat,1} + V_{GS,1} = 0.31+0-0.2415=68.5mV$
$V_{cm,in,max}$	$V_{DD} + V_{Dsat,7} + V_{GS,1} = 1.1-0.2415-0.2=685.5mV$
$V_{out,min}$	$V_{Dsat,6} = 0mV$
$V_{out,max}$	$V_{DD} + V_{Dsat,5} = 900mV$
P_{diss}	$V_{DD}(I_{BIAS} + I_{DC,S1} + I_{DC,S2}) = 1.1(2*360+2999)=4.1mW$



Design: Implementation of the OpAmp in MATLAB and LTspice (10 points):

Based on your handcalculations, create your OpAmp design in MATLAB.

Remarks:

- *Please make sure that all widths W are below 1mm*
- *Don't write your values calculated in the previous step as constants in MATLAB but write the full equation in your code.*
- *Values chosen as design choices should match between this table and the one from the handcalculations!*
- *Choose Vincm to be in the center of your input common-mode range. Do this in a 2-step process:*
  - *Use the center of the input common-mode range as calculated in the handcalculations for Vincm.*
  - Have MATLAB calculate the correct Vincm value using the calculated 'vdsat' fields of the transistors in MATLAB.

Fill out both tables depicted below. All values are to be determined in MATLAB.

Device sizes and bias point parameters according to MATLAB

Device	W [µm]	L [nm]	I_{ds} [µA]	V_{OV} [V]	g_m [µS]	g_{ds} [µS]	g_m/g_{ds} [-]	$V_{ds,sat}$ [V]	V_{ds} [V]
M_{p1}	460.3	500	180	-0.0018	3804.8	93.46	40.71	-0.06	-0.3645
M_{p2}	460.3	500	180	-0.0018	3804.8	93.46	40.71	-0.06	-0.3645
M_{n3}	24.15	500	180	0.1965	2493.8	85.65	29.11	0.1157	0.3173
M_{n4}	24.15	500	180	0.1965	2493.8	85.65	29.11	0.1157	0.3173
M_{p5}	972.75	1000	2992.7	-0.200	25233.7	667.95	37.779	-0.1752	-0.55
M_{n6}	436.43	100	2992.7	-0.7012e-6	58644	2635.5	22.25	0.0578	0.55
M_{p7}	120.97	1000	360	-0.2012	3034.4	109.28	27.77	-0.1762	-0.4182
M_{p8}	120.97	1000	360	-0.2012	3034.4	109.28	27.77	-0.1761	-0.4182

Device	Units	Value
C_M	pF	5
R_M	Ω	85.26
I_{BIAS}	A	360µ

Based on the parameters filled in the table above, design your OpAmp in LTspice. After completion, please insert your final and entire LTspice-netlist after the appendix (i.e. at the end of this reportdocument).

Experiment (10 points):

In this section you will simulate the following:

A. Frequency Response in MATLAB and LTspice
B. Noise contribution in LTspice
C. Linearity in LTspice

Note on how to present graphs in general:

- *When you make a graph, put labels on every axis to make clear what you are showing. Also, show the units!*
- *When you plot multiple curves on one graph, add a legend.*

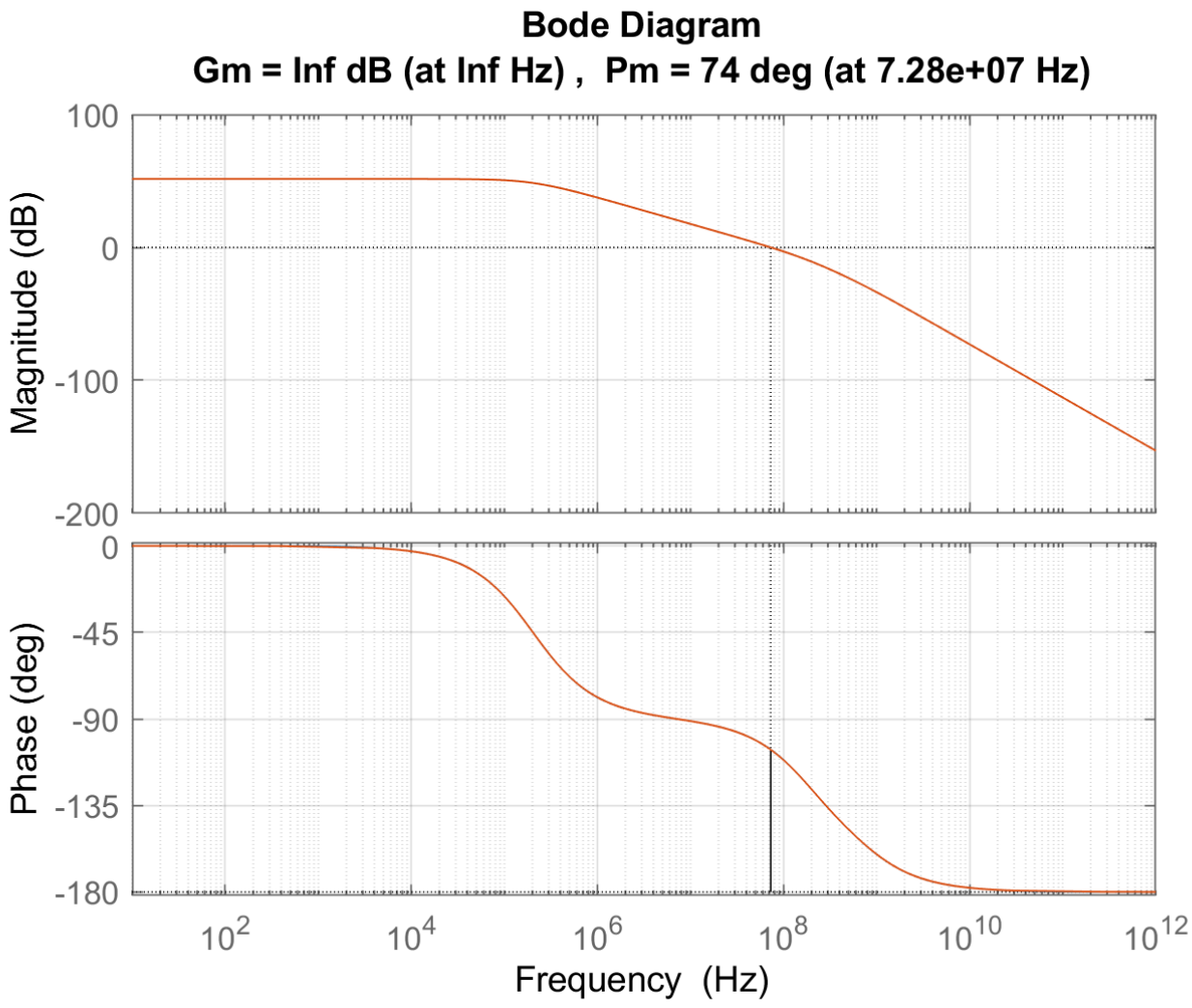
A. Frequency Response in MATLAB and LTspice

Note on how to present graphs in this sub-section:

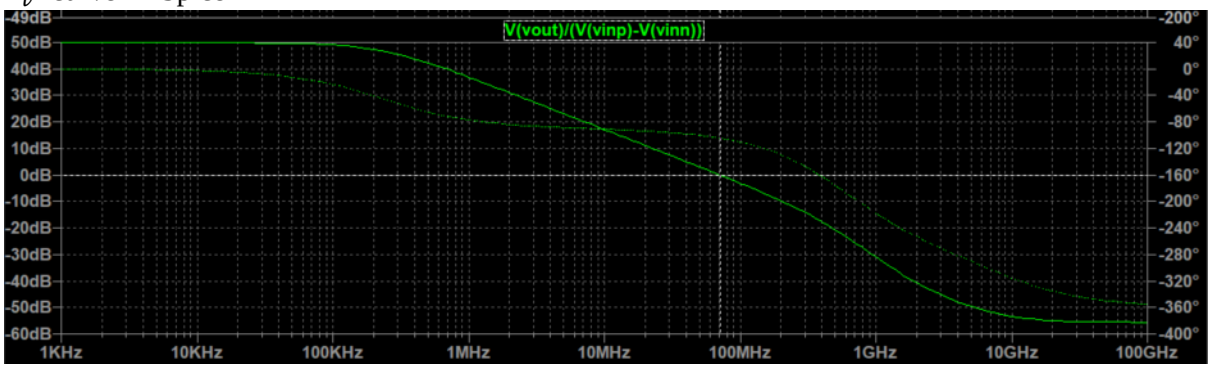
- ***For magnitude plots use dB (linear scale) vs. Hz (logarithmic scale)***
- ***for phase plots use ° (linear scale) vs. Hz (logarithmic scale).***

Simulate A_v (small voltage gain) with C_M and R_M in MATLAB and LTspice. Then, paste both A_v -curves in separate plots below such that the difference between MATLAB and LTspice can be clearly seen. In addition to that, indicate the resulting phase margin (PM) in both plots.

Plots:



A_v- curve LTSpice



Simulator	PM (deg)
MATLAB	74
LTspice	88

Explain, analyse and interpret the results above. Furthermore, if you observe significant differences between MATLAB and LTspice, explain why.

Simplified small-signal model lacks parasitics

In MATLAB you built only a first- or second-order transfer function. That representation omits MOSFET parasitic capacitances Cgd and Cdb, as well as interconnect inductances L. LTspice, on the other hand, uses transistor models that include those parasitic elements. The extra pole(s) they introduce add several degrees of additional phase lag at high frequencies, so the SPICE phase-margin comes out larger.

Device parameters taken directly from the data sheet, without bias-point effects

Your MATLAB model inserts DC parameters (gm, ro, Cox , etc.) straight from the data sheet and ignores how they vary with temperature, bias voltage, and over-drive. LTspice employs full BSIM/PSP transistor models that calculate those parameters at the exact operating point found by the simulator. The bias-dependent values differ slightly from the nominal data-sheet numbers, producing a small shift in gain and phase and hence a different phase-margin reading.

Simulate A_v in LTspice for the following cases:

a. No compensation network ($C_M = R_M = 0$).

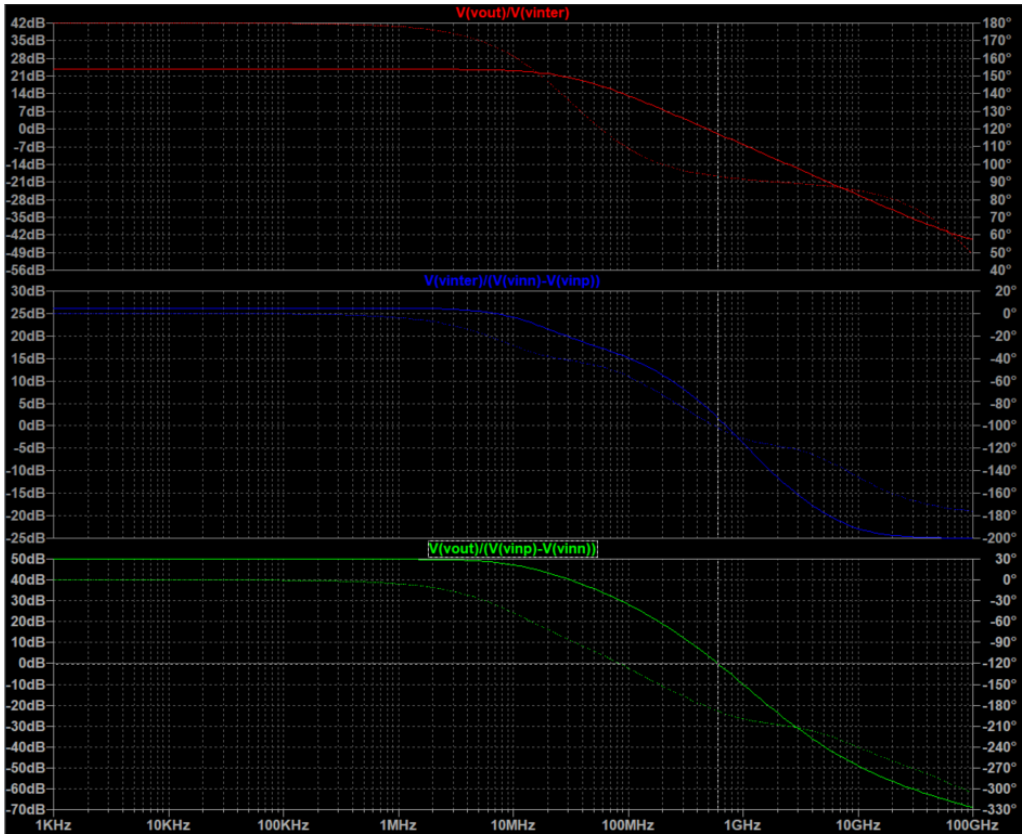
b. With compensation capacitor but no compensation resistor ($R_M = 0$).

c. With both the compensation capacitor and the compensation resistor.

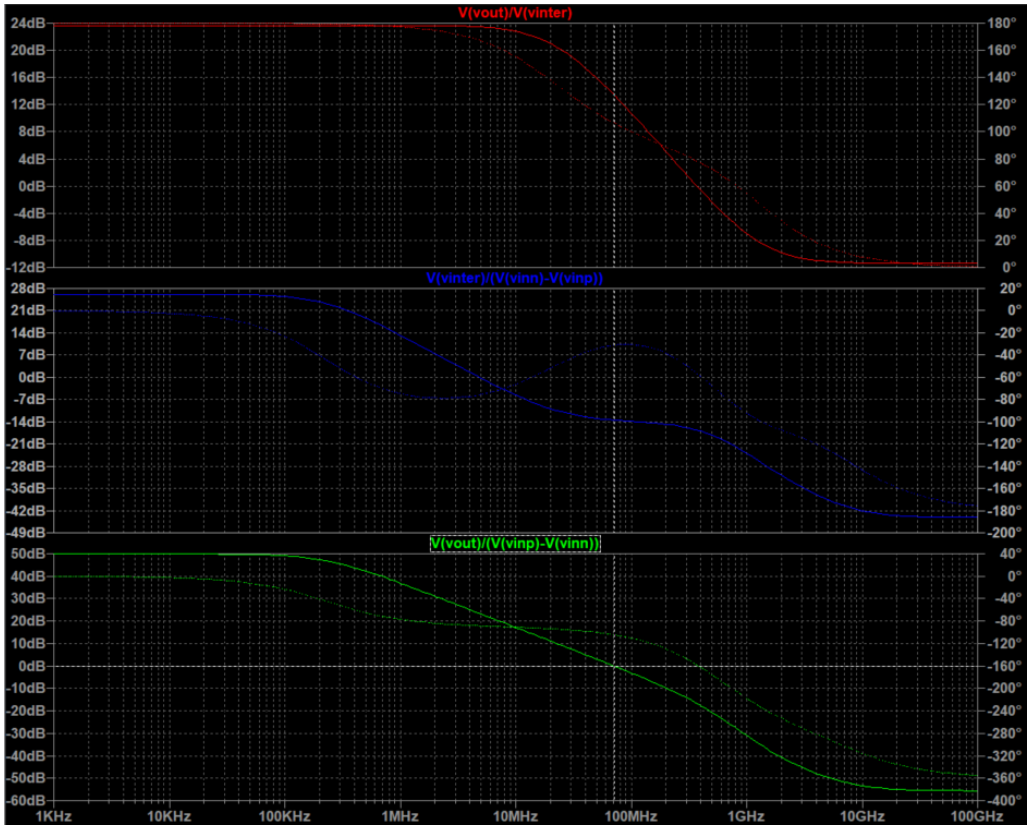
Then, show all 3 cases in seperate plots such that the differences are clearly visible. Furthermore, indicate on each plot the resulting PM.

Plot:

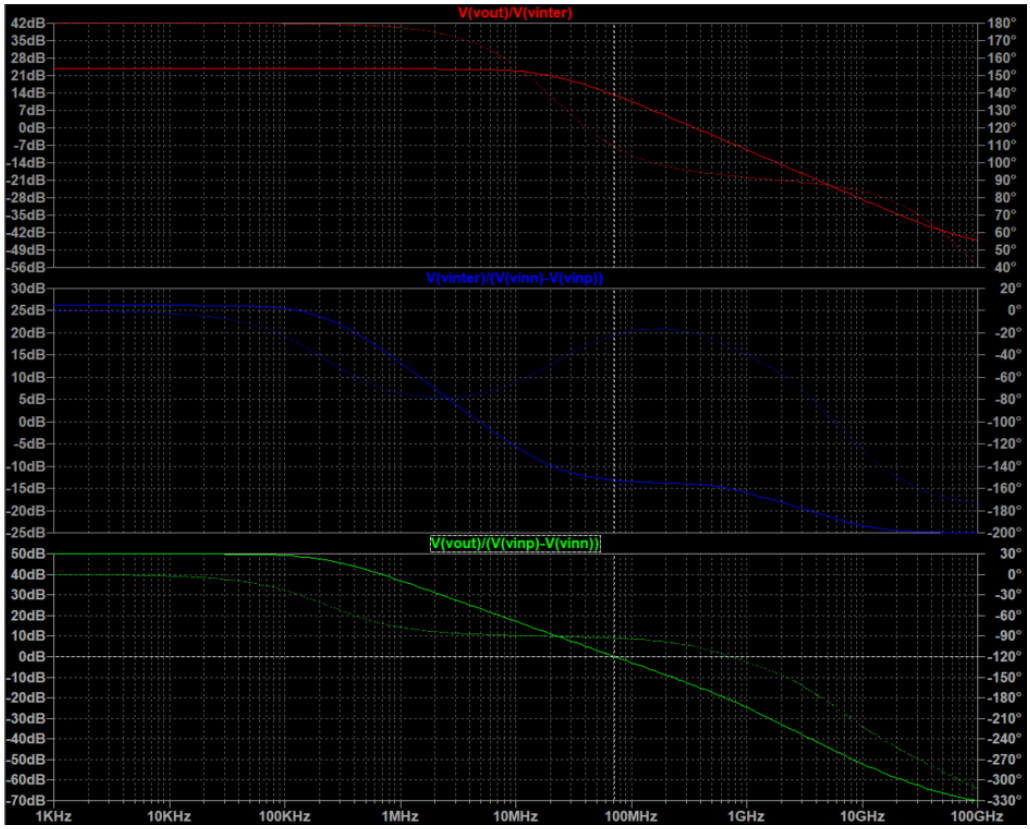
No compensation network



With compensation capacitor but no compensation resistor ($R_M = 0$)



With both the compensation capacitor and the compensation resistor.



Case	PM [deg]
No compensation network	8.4
No compensation resistor	76
With both compensation components	88

Explain, analyse and interpret the results above (for each case, explain the effect it has on the plots and PM.

With **Cm = 5 pF** (Rm = 0), the Miller capacitor between **vinter** and **vout** separates the originally adjacent first- and second-stage poles. By the Miller effect, the capacitance seen at the first-stage output becomes previous capacitor +Cm(1+|Av 2|) Since A2≈−17 (i.e., |A2|≈17), the reflected term is Cm(1+|A2|)≈5 pF×18≈90 pF, which is much larger than the **20 pF** load. As a result, the **first-stage pole** moves to a much lower frequency (the 0-dB point of V(vinter)/(V(vinn)−V(vinp)) shifts far to the left), while the low-frequency gains of the stages are essentially unchanged.

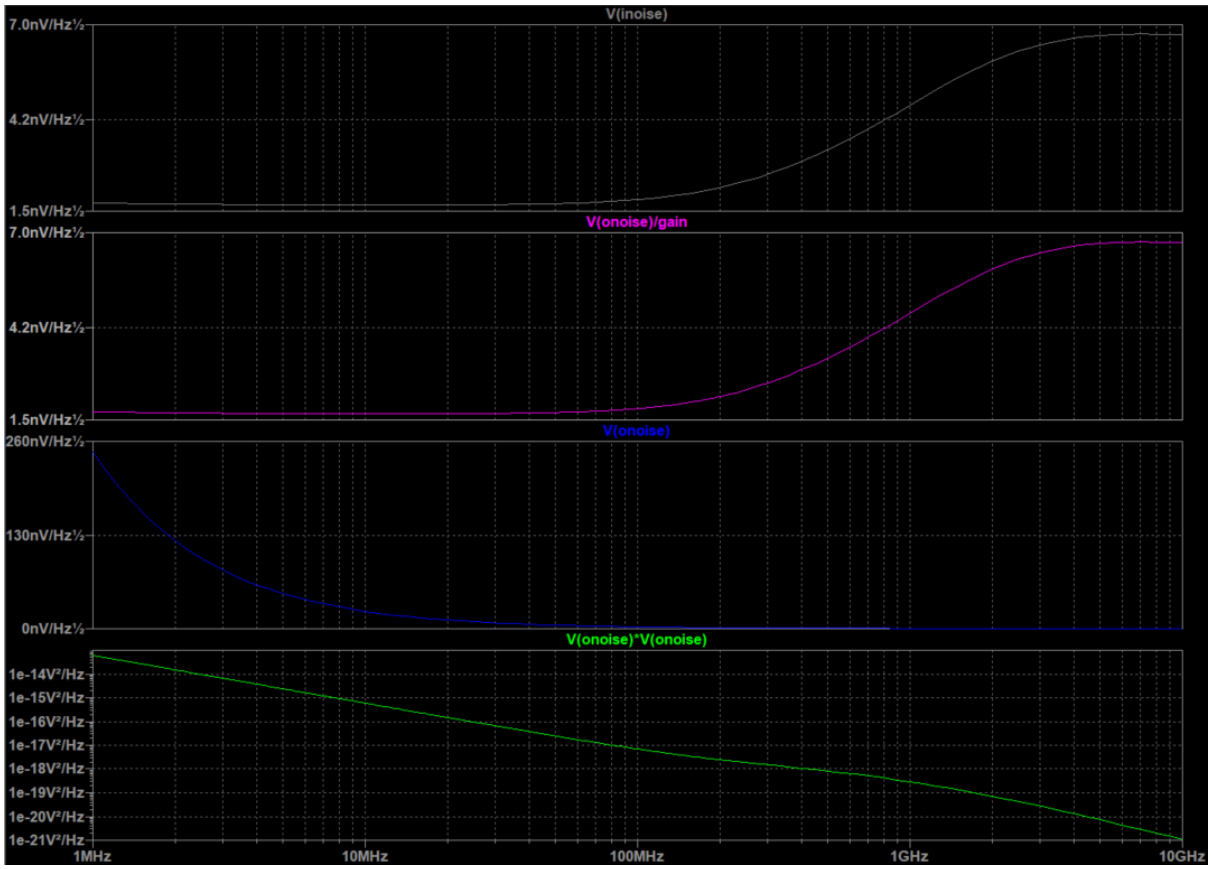
Before adding the Miller capacitor, the dominant pole could be approximated at the output stage, pout≈gm6/Cload. After adding **Cm**, the dominant pole relocates to the first-stage output node **vinter** due to the large Miller-amplified capacitance.

In our case, the resistor have some improve for phase, but not much help. Actually when we only have the capasitor in the circuit, the phase already arrive 70 degree, which means the system already stable.

B. Noise contribution in LTspice

Simulate the output-referred noise voltage power density $\overline{dv_{out,eq}^2}/\Delta f$ over an appropriate frequency range in LTspice. Then, insert the $\overline{dv_{out,eq}^2}/\Delta f$-plot below (use a logarithmic plot).

Plots:



Explain, analyse and interpret the results above (what regions can you see in the plot, and why?).

V(onoise) represents the output noise spectral density, which is directly affected by the amplifier's open-loop gain. At high frequencies (beyond the GBW), the open-loop gain drops sharply, so the output noise decreases rapidly.

V(onoise)/gain is effectively the output noise referred back to the input:

$\frac{\text{V(onoise)}}{\text{gain}} = \frac{\text{output-referred noise voltage spectral density}}{\text{open-loop gain}|Av|}$=V(inoise)

Since the open-loop gain |Av| becomes very small at high frequencies (much less than 1), dividing by this small number amplifies the result, making the curve appear to rise sharply.

At frequencies beyond the unity-gain bandwidth (fGBW=120MHz), the amplifier's open-loop gain decreases significantly, causing the noise transfer function to diminish dramatically. Consequently, the output noise amplitude spectral density (V(onoise)) sharply decreases at higher frequencies.

The green trace is the **output noise power spectral density**, Sv(f)=|Vonoise|2/Hz, simply the square of the blue "V(onoise)" curve plotted on a log–log scale. Past the dominant pole it falls with an approximately 1/f slope, which is the signature of a single-pole low-pass shaping white noise.

Simulate the input-referred noise spectral density by using the .NOISE option in LTspice. Also, by using LTspice, determine the total output integrated noise of the OpAmp. For that, take the integration bandwidth from 10 kHz to $10 \cdot f_{GBW}$. Furthermore, copy and paste the top 9 contributing elements using "View->SPICE Error Log" option and insert that list below.

```
out_totnvrms: INTEG(v(onoise))=0.00024637 FROM 1e+06 TO 1e+10
out_mp1nvrms: INTEG(v(mp1))=0.000135594 FROM 1e+06 TO 1e+10
out_mp2nvrms: INTEG(v(mp2))=0.000135643 FROM 1e+06 TO 1e+10
out_mp7nvrms: INTEG(v(mp7))=4.7086e-06 FROM 1e+06 TO 1e+10
out_mp8nvrms: INTEG(v(mp8))=5.68897e-06 FROM 1e+06 TO 1e+10
out_mp5nvrms: INTEG(v(mp5))=8.00629e-06 FROM 1e+06 TO 1e+10
out_mn6nvrms: INTEG(v(mn6))=1.25633e-05 FROM 1e+06 TO 1e+10
out_mn4nvrms: INTEG(v(mn4))=0.000111089 FROM 1e+06 TO 1e+10
out_mn3nvrms: INTEG(v(mn3))=0.000103252 FROM 1e+06 TO 1e+10
out_rmnvrms: INTEG(v(rm))=2.52007e-05 FROM 1e+06 TO 1e+10
```

Explain, analyse and interpret the results in above (what regions can you see in the plot, and why? + why is every transistor ranked the way it is for noise contribution).

To get the RMS output noise over a band [f1,f2], integrate the PSD and take the square root:

$$v_{n,\mathrm{rms}} = \sqrt{\int_{f_1}^{f_2} S_v(f)\, df}$$

1. MP1 & MP2 – Differential input pair
- Thermal noise current of the differential pair directly modulates the input nodes (vinp/vinn).
- The noise is then amplified by both gain stages, so it dominates the overall RMS noise (≈ 135.6 µV rms each).

2. MN4 & MN3 – Second-stage gain transistors
- Their channel noise appears almost directly at the output because these devices sit in the high-gain common-source second stage and are resistively connected to vout.
- Large g_m and only a few poles in the path give the next-largest contribution (≈ 110 µV and 100 µV rms).

3. Rm (62.7Ω) – Series nulling resistor
- Own 4 kT R voltage noise is in series between vinter and vout, so it is only weakly attenuated.
- Adds ≈ 25µV rms to the total output noise.

4. MN6 – Second-stage current-source NMOS
- Noise current flows into the load capacitance and couples to vout through a short path, but g_m is modest.
- Contribution ≈ 12 µV rms.

5. MP5 – Second-stage PMOS load
- Load device noise reaches vout in a way similar to MN6, but division through current-splitting and parasitic

capacitance makes it smaller.
- Contribution ≈ 8 µV rms.

6. MP7 & MP8 – First-stage current-mirror devices
- Noise is injected from heavily filter by the Miller capacitor C_m before being re-amplified by the second stage.
- Only a small fraction reaches the output (≈ 5.7µV and 4.7 µV rms).

Conclusion:

Thermal noise of the input differential pair is the primary noise bottleneck. 1/f noise usually in the low frequency.

The second-stage common-source devices are the next most significant source.

The compensation resistor Rm is non-negligible and can limit further noise reduction.

C. Linearity

Explain why gain compression occurs when you increase the input voltage.

. When the input signal gets big, one MOSFET in the input pair turns nearly off while the other carries almost all the current. Because the two devices no longer share the current, their combined transconductance drops to about half, so the first-stage gain falls.

2. A large output swing pushes the output node very close to the supply rails. With so little voltage left across the output transistors, they leave their normal operating region, their resistance becomes small, and the overall voltage gain collapses.

Assume an input voltage is applied such that the full voltage swing (as calculated) is utilized. Does increasing the input voltage still increase the output voltage at this point? Explain.

When the applied input already forces **vout** to the rail-to-rail limits (≈ VDD − VDS,sat and VSS + VDS,sat), the output transistors leave their linear region, gmrog_m r_ogmro collapses, and the stage can generate no additional voltage swing. Pushing the input higher merely drives the devices deeper into saturation or slew-rate limiting, increasing distortion but **not** the output amplitude.

As we keep increasing the signal, the intermediate node and the output are pushed to within only a few hundred millivolts of the rails. The top-side and bottom-side transistors no longer have enough drain-source voltage to stay in saturation, so their small-signal resistance collapses. Since stage gain is Av=gm×ro, transistors **MN4/MN6** and **MP5** also lack sufficient VDS; as a result, their output resistance ro drops sharply.

BONUS: I didn't set up my large-signal simulation parameters correctly and my gain vs Vin plot is first rising, then stays flat and finally it starts dropping. What would I have expected to see and what could have gone wrong?

Conclusion (10 points):

14) Conclude your experiment by filling the editable fields in the performance table depicted below

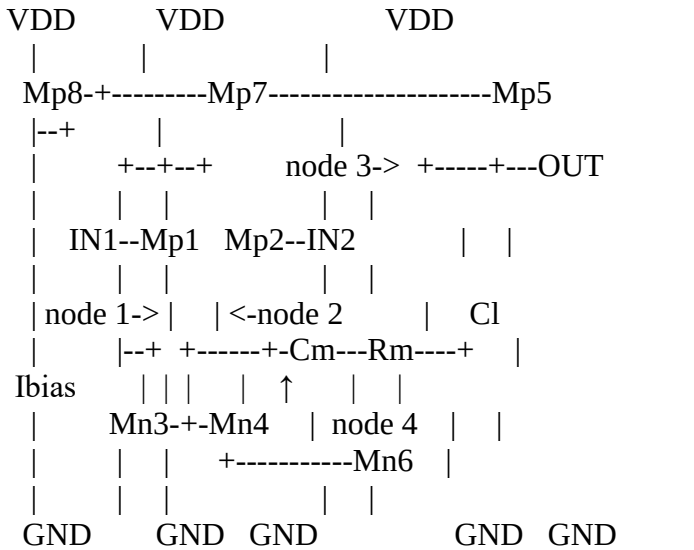
Metric	Units	Specification	from hand-calculations	MATLAB	LTspice
DC gain	magnitude	316.228	502.128	377.11	282
DC gain	dB	50	54	51.53	50
Gain-Bandwidth frequency f_{GBW}	MHz	120	113.538	114.7	75
Dominant pole frequency f_{dom}	kHz	379.47	226.114	304.029	240
PM	°	> 70	76	74	88
$V_{in,cm,max} - V_{in,cm,min}$	V		0.617	0.546	0.617
$V_{out,max} - V_{out,min}$	V	> 0.7	0.9	0.867	0.87
P_{diss}	mW		0.41	0.41	4.25
$\overline{V_{out.eq}^2}$(output-referred noise)	V_{rms}				246*10^-3

Comment about the deviations between hand calculation, MATLAB and LTspice values found above. Conclude, what the causes of such deviations are.

If you have results which are not passing the specifications: conclude for each of those result what the cause is and what you need to do in order to make it pass, if you were repeating this experiment.

Appendix:

Previously, you were expected to include your MATLAB code and LTspice netlist here, but now we changed it so that you must upload them in the Canvas assignment. Read the description of the assignment for details on exactly what files are needed!




```
=====================================
=     Transistors in saturation     =
=====================================

Mp1:Success
Mp2:Success
Mn3:Success
Mn4:Success
Mp5:Success
Mn6:Success
Mp7:Success
Mp8:Success

=====================================
=    Sizes and operating points     =
=====================================
IBIAS  =   0.36mA
Rm     =  85.26 Ohm
Cm     =   5 pF

=====================================
=       Performance                 =
=====================================

metrik              result
Vin,cm,min [mV]       136
Vin,cm,max [mV]       682
Vout,cm,min [mV]      58
Vout,cm,max [mV]      925
Pdiss [mW]            4.1
AvTotdB =            51.5293
fGBW [MHz]            114.7
```